1) Evaluate the definite integral $\int_0^{\pi} (1+\cos x)\cdot dx$

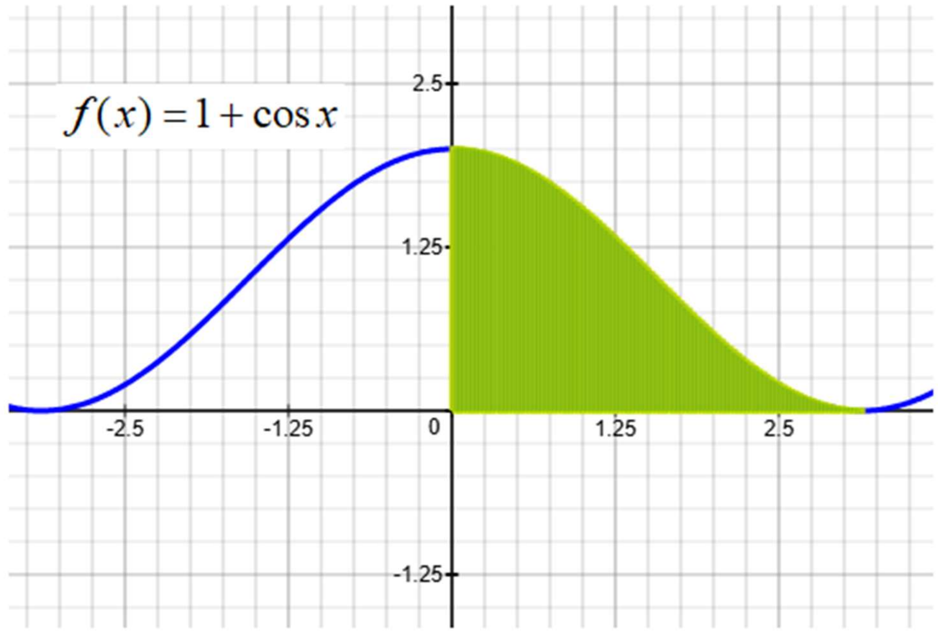
a) Antiderivate $F(x) = \int (1+\cos x)\cdot dx =$ __?__

b) $F(\pi) =$ __?__

c) $F(0) =$ __?__

d) $\int_0^{\pi} (1+\cos x)\cdot dx = F(\pi) - F(0) =$ __?__

e) Area of shaded region $=$ __?__



2) Evaluate the definite integral $\int_{-\pi/6}^{\pi/6} \sec^2 x \cdot dx$

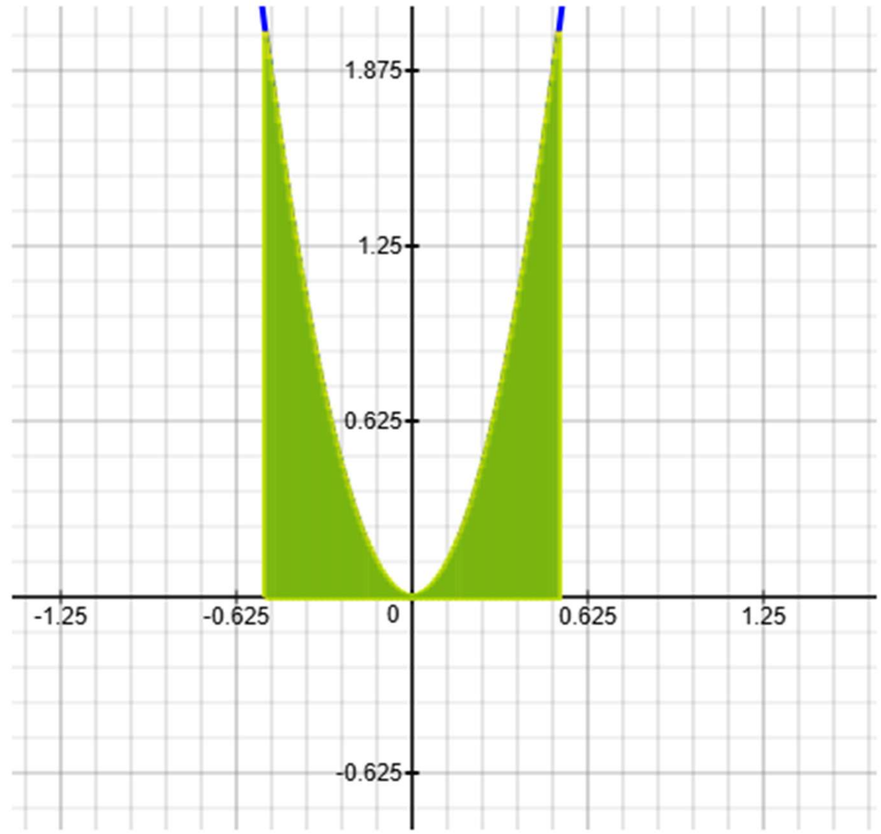
a) Antiderivate $F(x) = \int (\sec^2 x) \cdot dx =$ ___?___

b) $F(\pi/6) =$ ___?___

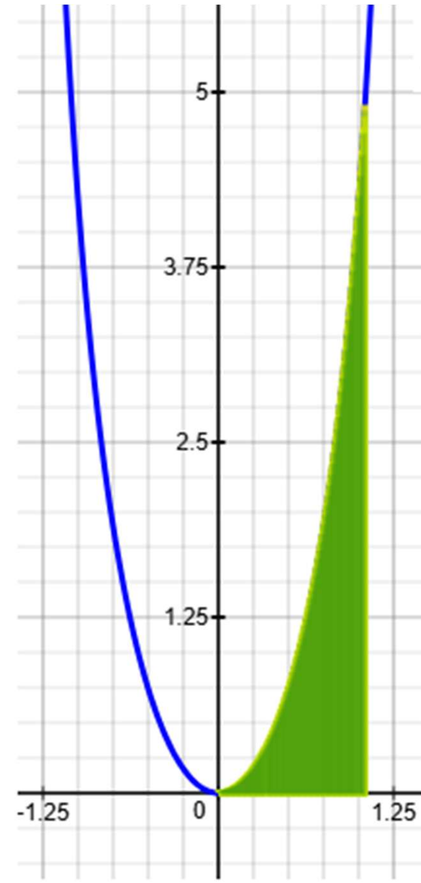
c) $F(-\pi/6) =$ ___?___

d) $\int_{-\pi/6}^{\pi/6} \sec^2 x \cdot dx = F(\pi/6) - F(-\pi/6) =$ ___?___

e) Area of shaded region $=$ ___?___



3) Evaluate the definite integral $\int_0^{\pi/3} (\sec x \cdot \tan x) \cdot dx$



$f(x) = \sec x \cdot \tan x$

a) Antiderivate $F(x) = \int (\sec x \cdot \tan x) \cdot dx =$ ___?___

b) $F(\pi/3) =$ ___?___

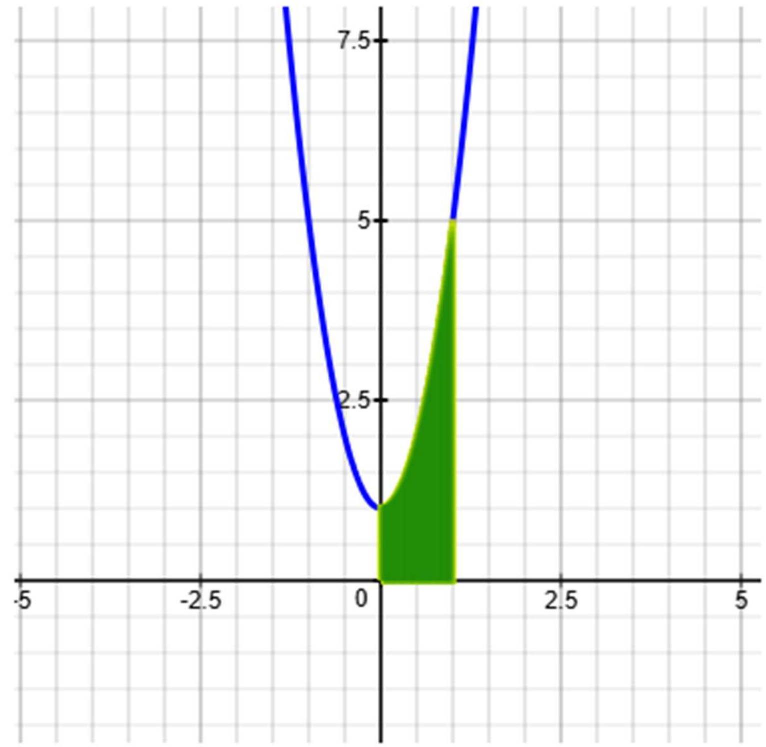
c) $F(0) =$ ___?___

d) $\int_0^{\pi/3} (\sec x \cdot \tan x) \cdot dx = F(\pi/3) - F(0) =$ ___?___

e) Area of shaded region $=$ ___?___

4) Find the area bounded by the following graphs:

$y = 4x^2 + 1;$ $\quad x = 0;$ $\quad x = 1;$ $\quad y = 0$



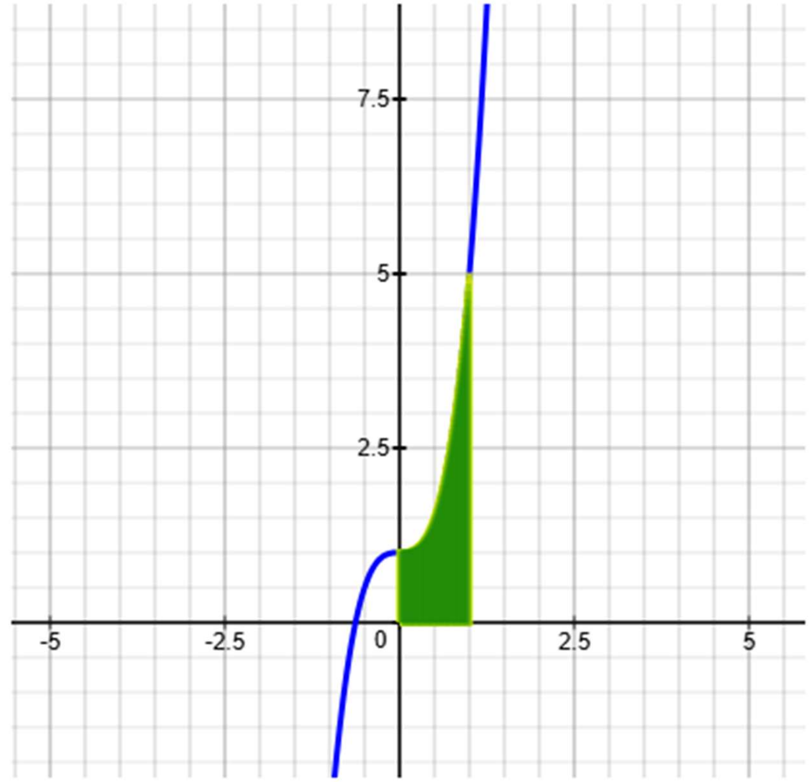
a) Set up definite integral representing shaded region.

$$\int_a^b f(x) \cdot dx = \underline{\quad ? \quad}$$

b) Area of shaded region.___?___

5) Find the area bounded by the following graphs:

$y = 4x^3 + 1;$ $\qquad x = 0;$ $\qquad x = 1;$ $\qquad y = 0$



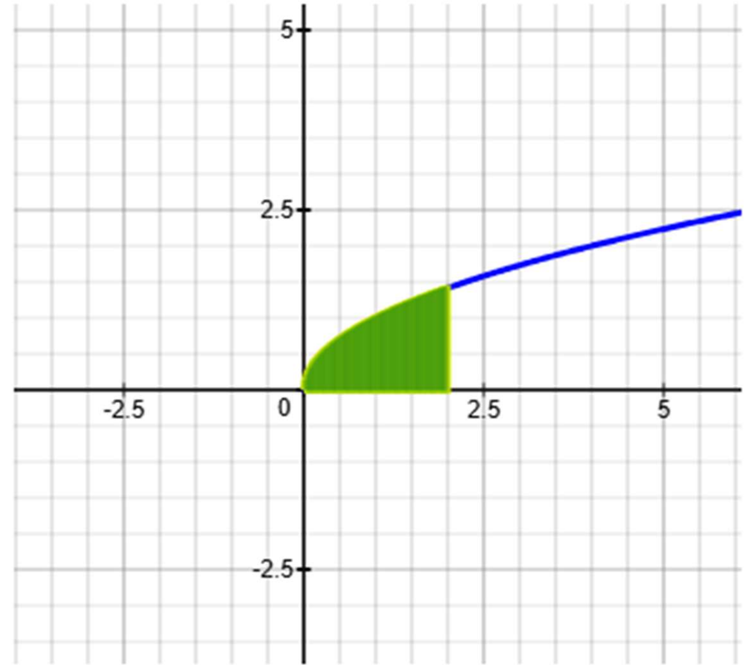
a) Set up definite integral representing shaded region.

$$\int_a^b f(x) \cdot dx = \underline{\quad ? \quad}$$

b) Area of shaded region.__?__

6) Find the area bounded by the following graphs:

$y = \sqrt{x}$; $\quad x = 0$; $\quad x = 2$; $\quad y = 0$



a) Set up definite integral representing shaded region.

$$\int_a^b f(x) \cdot dx = \underline{\quad ? \quad}$$

b) Area of shaded region.___?___